

APIA WHARF, Samoa

WMO: 917560

Lat: 13.826S

Lon: 171.761W

Elev: 3

StdP: 101.29

Time Zone: 13.00 (PCS)

Period: 98-11

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	22.4	N/A	N/A	N/A	N/A	N/A	N/A	8.5	27.3	7.7	27.3	1.4	180	0.324

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	5.5	31.2	N/A	30.6	N/A	30.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2.7	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 7.8	7.0	6.2	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.6	27.0	27.1	27.2	27.0	26.5	26.2	25.8	25.9	26.1	26.4	26.8	26.9		
	DBStd	0.90	0.85	0.86	0.87	0.66	0.78	0.81	0.73	0.70	0.66	0.68	0.73	0.76		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	6049	526	478	534	511	513	487	489	492	483	508	503	524		
	CDD18.3	3007	268	244	276	261	254	237	231	233	233	250	253	266		
	CDH23.3	28251	2632	2420	2748	2601	2336	2123	1899	2000	2089	2336	2452	2614		
	CDH26.7	6678	696	658	788	718	542	450	309	333	372	496	627	689		
(14) Wind	WSAvg	2.4	2.3	2.1	2.0	2.2	2.2	2.6	2.6	2.9	2.9	2.6	2.2	2.2		
(15) Precipitation	PrecAvg	2971	440	329	357	231	178	151	115	125	164	231	263	370		
	PrecMax	4074	960	935	696	528	365	327	292	307	593	579	727	933		
	PrecMin	1899	229	79	133	16	10	36	5	19	1	47	25	63		
	PrecStd	638	191	154	146	124	81	79	66	79	127	152	156	208		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.1	31.3	33.6	31.1	30.4	29.7	29.2	29.2	29.4	29.9	31.6	31.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	30.4	30.7	31.1	30.3	29.9	29.4	28.8	28.8	29.0	29.5	30.4	30.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	30.0	30.2	30.2	30.0	29.5	29.1	28.5	28.6	28.7	29.1	29.7	30.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	29.5	29.7	29.8	29.6	29.1	28.7	28.2	28.2	28.4	28.8	29.3	29.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	5.3	5.5	5.3	5.1	5.0	4.9	4.8	4.7	4.8	5.3	5.3		
		MCDBR	5.6	5.8	6.3	5.6	5.5	5.2	5.2	5.1	5.0	5.2	5.8	6.0		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.387	0.378	0.375	0.382	0.390	0.391	0.392	0.396	0.399	0.388	0.385	0.385		
	taud		2.625	2.659	2.670	2.644	2.604	2.586	2.573	2.556	2.552	2.597	2.618	2.626		
	Ebn at Noon		956	957	941	902	862	844	851	876	905	937	952	957		
	Edh at Noon		102	98	94	92	90	89	92	99	104	103	102	101		
(44) All-Sky Solar Radiation	RadAvg		5.71	5.71	5.73	5.15	4.54	4.34	4.49	5.20	5.82	6.00	5.89	5.81		
	RadStd		0.57	0.53	0.42	0.42	0.32	0.22	0.18	0.25	0.33	0.30	0.31	0.36		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page